Chapter 2 Review Problem Set

21 problems · 7 days · 3 problems per day

Scope

All ten Homework 2 exercises are included. Additional questions target Gaussian moments, correlation identities, finite-dimensional independence, and assumptions that matter in past papers. Each problem has its own page and handwriting space. A matched Chinese solution key uses the same numbering.

Source labels

Textbook problems from Homework 2 retain the existing English translation of the Chinese textbook. This is not an official translation. Questions labelled "Adapted from" restate or extend selected exam tasks; extra assumptions are explicit. Recall questions are newly written.

Required homework: 2.3, 2.13, 2.16, 2.19.
Optional homework: 2.1, 2.4, 2.7, 2.8, 2.11, 2.18.
Homework issued: 13 October 2026. Due: 27 October 2026.
This seven-day revision schedule does not change the homework deadline.

Seven-day plan and problem index

Day	Problems	Focus
1	1–3	Gaussian tools: Gaussian checklist; 2.19; 2.17
2	4–6	Gaussian applications: Square-law device; 2.16; 2.18
3	7–9	Phase and cross-correlation: Uniform phase; sampled sin/cos; 2.11
4	10–12	Correlation and independence: Identities; random line; signal independence
5	13–15	Finite-dimensional laws: 2.3; coin-toss exam; 2.1
6	16–18	Sequences and linear combinations: 2.4; 2.8; 2.7
7	19–21	Operations and assumptions: 2.13; 2.14; white-noise caveat

No.	Source	Textbook section
1	Recall / extension	2.4.1–2.4.3
2	Textbook 2.19 / Required	2.4.1
3	Textbook 2.17 / translated	2.4.1 / 2.4.3
4	Adapted from 2022–23 Q3	2.4.3
5	Textbook 2.16 / Required	2.4.1 / 2.4.3
6	Textbook 2.18 / Optional	2.4.1 / 2.4.3
7	Adapted from 2022–23 Q2(a,b)	2.2.1 / 2.5
8	Adapted from 2024–25 Q1	2.3.2 / 2.5
9	Textbook 2.11 / Optional	2.3.2 / 2.5
10	Recall / extension	2.1.3 / 2.3.3
11	Adapted from 2022–23 Q1	2.1.1–2.1.3
12	Recall / extension	2.3.1 / 2.5
13	Textbook 2.3 / Required	2.1.2 / 2.3.1
14	Adapted from 2025–26 Q1	2.1.2 / 2.3.1 / 2.5
15	Textbook 2.1 / Optional	2.1.1 / 2.2.1
16	Textbook 2.4 / Optional	2.2.2
17	Textbook 2.8 / Optional	2.2.2 / 2.1.3
18	Textbook 2.7 / Optional	2.4.2 / 2.1.3
19	Textbook 2.13 / Required	2.2.1 / 2.3.4
20	Textbook 2.14 / translated	2.3.4
21	Adapted from 2023–24 Q2(c)	2.4.3

Problem 1 Day 1

Gaussian checklist

(Recall / extension)

Textbook section: 2.4.1–2.4.3

Define a Gaussian process using its finite-dimensional distributions. For a Gaussian vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{C}$, state its characteristic function, its ordinary PDF (including the condition for its existence), and the law of $\boldsymbol{GX} + \boldsymbol{b}$.

Is marginal Gaussianity sufficient for joint Gaussianity? Give a counterexample. State when uncorrelatedness implies independence.

Working / Answer

Problem 2

Day 1

Joint Gaussian PDF and CF

(Textbook 2.19 / Required)

Textbook section: 2.4.1

Let the random vector $(X, Y) \sim N(\boldsymbol{\mu}, \boldsymbol{C})$, where

$$\boldsymbol{\mu} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \qquad \boldsymbol{C} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}.$$

Find the joint probability density function of (X, Y) and its characteristic function $\phi_{XY}(u, v)$.

Working / Answer

Problem 3

Day 1

Fourth-order Gaussian moment

(Textbook 2.17 / translated)

Textbook section: 2.4.1 / 2.4.3

Let $X(t)$ be a zero-mean Gaussian signal and put $X_k = X(t_k)$ for $k = 1, 2, 3, 4$. Prove that

$$E[X_1X_2X_3X_4] = E[X_1X_2]E[X_3X_4] + E[X_1X_3]E[X_2X_4] + E[X_1X_4]E[X_2X_3].$$

Working / Answer

Problem 4

Day 2

Gaussian signal through a square-law device

(Adapted from 2022–23 Q3)

Textbook section: 2.4.3

A zero-mean stationary Gaussian white noise $N(t)$ has PSD $N_0/2$, where $N_0 > 0$. It passes through an RC low-pass filter

$$H(j\omega) = \frac{1}{1 + j\omega RC}, \qquad R, C > 0,$$

whose output is $X(t)$. A second device produces $Y(t) = aX^2(t)$, where a is real.

(1) Find $R_X(\tau)$.

(2) Find $E[Y(t)]$, $R_Y(\tau)$ and $C_Y(\tau)$.

(3) Is $Y(t)$ Gaussian when $a \neq 0$?

The transfer function is supplied in place of the original circuit figure; the covariance and Gaussianity questions are extensions.

Working / Answer

Problem 5

Day 2

Gaussian sinusoid

(Textbook 2.16 / Required)

Textbook section: 2.4.1 / 2.4.3

Let the random process be

$$X(t) = A\cos\omega t + B\sin\omega t,$$

where ω is a constant and A and B are independent Gaussian random variables. Given that

$$E[A] = E[B] = 0, \qquad E[A^2] = E[B^2] = \sigma^2,$$

find the one-dimensional and two-dimensional probability density functions of $X(t)$.

Working / Answer

Problem 6

Day 2

Singular three-dimensional Gaussian law

(Textbook 2.18 / Optional)

Textbook section: 2.4.1 / 2.4.3

A Gaussian signal has mean and covariance function

$$m_X(t) = 2, \qquad C_X(t_1, t_2) = 8\cos(t_1 - t_2).$$

Write down its three-dimensional probability density function at the times

$$t_1 = 0, \qquad t_2 = 0.5, \qquad t_3 = 1.$$

Working / Answer

Problem 7

Day 3

Uniform random phase

(Adapted from 2022–23 Q2(a,b))

Textbook section: 2.2.1 / 2.5

Let $\eta(t) = \xi(t)\cos(\omega_0 t + \Theta)$, where $\xi(t)$ is a real WSS process with finite second moment and $\Theta \sim U[0, 2\pi]$ is independent of the entire process. Find the mean and autocorrelation of $\eta(t)$, and determine whether it is WSS. Show the phase averages explicitly. No PSD calculation is required.

Working / Answer

Problem 8

Day 3

Orthogonal but dependent signals

(Adapted from 2024–25 Q1)

Textbook section: 2.3.2 / 2.5

Let

$$X[n] = \sin(2\pi n + \Phi), \qquad Y[n] = \cos(2\pi n + \Phi), \qquad n \in \mathbb{Z},$$

where $\Phi \sim U(0, \pi)$.

(1) Find $R_{XY}(m, n)$ and the two means.

(2) Are the signals orthogonal, uncorrelated, or statistically independent? Justify each answer.

(3) Are they jointly stationary?

Working / Answer

Problem 9

Day 3

Pairwise versus mutual independence

(Textbook 2.11 / Optional)

Textbook section: 2.3.2 / 2.5

Consider the two random signals

$$X(t) = A\sin(\omega t + \Theta), \qquad Y(t) = B\cos(\omega t + \Theta),$$

where A and B are random variables with unspecified distributions, Θ is uniformly distributed on $[0, 2\pi]$, and ω is a constant. The random variables A, B, and Θ are pairwise statistically independent.

(1) Find the cross-correlation function $R_{XY}(t_1, t_2)$ of the two random signals.

(2) Discuss their orthogonality, uncorrelatedness, and statistical independence.

Working / Answer

Problem 10 Day 4

Correlation identities

(Recall / extension)

Textbook section: 2.1.3 / 2.3.3

For a real process $X(t)$ with finite second moments:

(1) Define m_X, R_X and C_X, and relate $R_X(t,t)$ to variance.

(2) Prove that every sampled covariance matrix is positive semidefinite, and that $|C_X(s,t)|^2 \le C_X(s,s)C_X(t,t)$.

(3) For $Y(t) = X(t+a) - X(t)$, expand $R_Y(s,t)$ without assuming stationarity.

Working / Answer

Problem 11

Day 4

A random straight line

(Adapted from 2022–23 Q1)

Textbook section: 2.1.1–2.1.3

Let $X(t) = U + t$ for $-2 \le t \le 2$, where $U \sim U[0, \pi]$. Sketch two sample functions, find the characteristic function of $X(t)$, and compute its mean and autocovariance. Explain whether samples at distinct times are independent.

Working / Answer

Problem 12

Day 4

Independence at the process level

(Recall / extension)

Textbook section: 2.3.1 / 2.5

State the finite-dimensional definition of independence of two random signals. Does independence of $X(t)$ and $Y(t)$ at every equal time imply independence of the two signals? Use an iid sequence $\{Z_n\}$ of fair signs and $X_n = Z_n$, $Y_n = Z_{n-1}$ to answer. Also state the criterion for two jointly Gaussian signals to be independent.

Working / Answer

Problem 13 Day 5

One toss, two observation times

(Textbook 2.3 / Required)

Textbook section: 2.1.2 / 2.3.1

A coin is tossed, and a random process is defined by

$$X(t) = \begin{cases} \cos \pi t, & \text{if heads occurs,} \\ 2t, & \text{if tails occurs.} \end{cases}$$

Heads and tails are equally likely. Find:

(1) The one-dimensional distribution functions $F_X(x, 1/2)$ and $F_X(x, 1)$ of $X(t)$;

(2) The two-dimensional distribution function $F_X(x_1, x_2; 1/2, 1)$ of $X(t)$;

(3) Sketch the graphs of the above distribution functions.

Working / Answer

Problem 14 Day 5

One toss versus fresh tosses

(Adapted from 2025–26 Q1)

Textbook section: 2.1.2 / 2.3.1 / 2.5

A fair coin is tossed once. Define $X(t) = \sin(2\pi t)$ for heads and $X(t) = t^2$ for tails.

(1) Find the generalized PDF of $X(2)$.

(2) Find the joint generalized PDF of $(X(1), X(2))$.

(3) Find $C_X(s,t)$.

(4) Now define $Y[n] = \sin(2\pi n)$ or n^2 by a fresh independent fair toss at each integer n. Find the joint generalized PDF of $(Y[1], Y[2])$.

Working / Answer

Problem 15

Day 5

Random amplitude and a degenerate time

(Textbook 2.1 / Optional)

Textbook section: 2.1.1 / 2.2.1

Consider the sinusoidal random signal $X(t) = V\cos\omega t$, where $0 \le t < \infty$, ω is a constant, and V is uniformly distributed on $[0, 1]$.

(1) Sketch two sample functions of this process.

(2) Determine the probability density function of the random variable $X(t_i)$ at each of the times

$$t_i = 0, \quad \frac{\pi}{4\omega}, \quad \frac{3\pi}{4\omega}, \quad \frac{\pi}{\omega}.$$

(3) Find the probability density function of $X(t_0)$ when $t_0 = \pi/(2\omega)$.

Working / Answer

Problem 16 Day 6

Bernoulli words

(Textbook 2.4 / Optional)

Textbook section: 2.2.2

Suppose that the binary data sequence $\{B(n), n = 1, 2, 3, \ldots\}$ is a Bernoulli random sequence. Each bit corresponds to a random variable $B(n)$, with

$$P[B(n) = 0] = 0.2, \qquad P[B(n) = 1] = 0.8.$$

(1) What is the probability that four consecutive bits form the word {1011}?

(2) What is the mean word formed by four consecutive bits?

(3) Which word formed by four consecutive bits has the highest probability?

Working / Answer

Problem 17

Day 6

A fair-sign sequence

(Textbook 2.8 / Optional)

Textbook section: 2.2.2 / 2.1.3

Suppose that the Bernoulli sequence $\{I_n, n = 1, 2, \ldots\}$ takes the values -1 and $+1$ with equal probabilities. Find its one-dimensional probability density function, mean, and covariance function.

Working / Answer

Problem 18

Day 6

Gaussian straight-line process

(Textbook 2.7 / Optional)

Textbook section: 2.4.2 / 2.1.3

The position of a moving particle is described by the straight-line process

$$X(t) = Vt + X_0,$$

where $V \sim N(1, 1)$ and $X_0 \sim N(0, 2)$ are independent.

Find the one-dimensional probability density function, mean, and variance of the random variable $X(t)$ at time t.

Working / Answer

Problem 19

Day 7

Random voltage across a parallel RC circuit

(Textbook 2.13 / Required)

Textbook section: 2.2.1 / 2.3.4

Consider the sinusoidal voltage signal

$$X(t) = A\cos(\omega t + \Theta),$$

where ω is a constant, $A \sim U(-1, +1)$, and $\Theta \sim U(-\pi, +\pi)$. The random variables A and Θ are independent. If this voltage is applied across a parallel RC circuit, find the total current signal and its mean-square value.

Working / Answer

Problem 20 Day 7

Integration of a random signal

(Textbook 2.14 / translated)

Textbook section: 2.3.4

For the integral of a random signal,

$$Y(t) = \int_0^t X(u)\,du, \qquad t \geq 0,$$

find expressions for its mean and autocorrelation function. State sufficient conditions that justify interchanging integration and expectation.

Working / Answer

Problem 21

Day 7

White noise is not a Gaussian assumption

(Adapted from 2023–24 Q2(c))

Textbook section: 2.4.3

A zero-mean white noise $W(t)$ with PSD $N_0/2$ passes through $h(t) = e^{-t}u(t)$ to produce $X(t)$. Let $Y(t) = X(t) - X(t-T)$, with $T > 0$ and $N_0 > 0$. You may use $R_X(\tau) = N_0 e^{-|\tau|}/4$.

(1) Do the stated assumptions imply that $Y(t)$ is Gaussian?

(2) Under the additional assumption that W is Gaussian white noise, find the one-dimensional PDF of $Y(t)$.

(3) Describe the case $T = 0$.

The extra Gaussian assumption in (2) is explicit; it was not stated in the original exam stem.

Working / Answer